

MANAV VIPIN NAIR

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ECE undergraduate (NIT Trichy) building full-stack and applied-AI systems — graph databases, LLM agents, real-time web apps.

EDUCATION

National Institute of Technology, Trichy

B.Tech – Electronics and Communication Engineering | CGPA: 8.47

Trichy, India

2024 – Present

Devi Academy Senior Secondary School

Class XII (CBSE) | 95.4%

Chennai, India

2024

PSBB Millennium, Gerugambakkam

Class X (CBSE) | 98.2%

Chennai, India

2022

EXPERIENCE

Software Engineering Intern

May 2026 – Jun 2026

NestCyber Campus

Remote

- Built Sentinel, a graph-based fraud-detection and agentic investigation platform, as the sole developer.
- Worked in a mentor-reviewed, issue-driven workflow with an industry-standard Git process — feature branches, pull requests, code review, and CI.

Deputy Manager, WebOps — Probe '25

Dec 2025 – Present

NIT Trichy – ECE Department Symposium

Trichy, India

- Contributed to full-stack development and maintenance of the official symposium website.
- Managed real-time updates and optimised the interface for a seamless participant experience.

Web Developer

Jan 2026 – Present

NIT Trichy – Counselling, Guidance & Welfare Cell

Trichy, India

- Designed and developed the official cell website in React — a responsive, accessible interface for students to access counselling resources and guidance information.

Manager, Events Team

Apr 2025 – Present

NIT Trichy – Festember '25

Trichy, India

- Spearheaded event ideation and end-to-end event management, introducing novel activities and ensuring smooth coordination between sub-teams.

PROJECTS

Sentinel

May 2026 – Jun 2026

Neo4j, Qdrant, FastAPI, React, TypeScript, Docker, LangGraph, Gemini

- Built a graph-based fraud-detection platform modelling money movement as a graph of 200+ users and 1,000+ transactions linked to shared IPs and devices, surfacing fraud rings through shared-entity structure rather than pre-set labels.
- Architected a multi-service system — Neo4j (graph DB), Qdrant (vector DB), a FastAPI investigation service, and a React + TypeScript dashboard — with the data layer orchestrated via Docker Compose.
- Modelled transactions as first-class nodes to enable multi-hop traversal and cycle detection, catching circular-payment, rapid-transaction, and shared-infrastructure fraud patterns.
- Built a weighted risk-scoring engine over graph signals, producing per-user scores and tiers (LOW/MEDIUM/HIGH/CRITICAL), with Qdrant cosine-similarity search surfacing users with similar risk profiles.
- Built a 5-node LangGraph workflow (Google Gemini) that retrieves the flagged subgraph and similar cases, grounding LLM-generated forensic summaries in the graph data (RAG-style).

StudySync

Nov 2025 – Dec 2025

MERN, Socket.io, Redis, TensorFlow.js, JWT

- Built a full-stack real-time study-room collaboration platform with secure Bcrypt/JWT authentication.
- Engineered real-time state sync via WebSockets (Socket.io) scaled across multiple Node.js instances with a Redis Pub/Sub adapter to remove single-server bottlenecks.
- Implemented persistent chat with cursor-based pagination and compound indexing on roomId/createdAt for infinite scroll without client-side memory overload.
- Integrated a client-side TensorFlow.js (BlazeFace) model for a privacy-first, in-browser study-proctoring timer.

Probe '25 Portal

Dec 2025

React, TypeScript, Tailwind CSS, Supabase, PostgreSQL, Drizzle ORM

- Built the official full-stack event & accommodation management portal for the ECE Department symposium at NIT Trichy.

- Implemented a mobile-responsive frontend from Figma designs; architected a Supabase (PostgreSQL) backend with OAuth authentication and type-safe Drizzle ORM access.

Titan's Circuit

May 2025

Vanilla JavaScript, HTML, CSS

- Built a turn-based strategy game on a three-ring concentric hexagonal graph (18 nodes, weighted edges) using only vanilla JS, HTML, and CSS — no Canvas or frameworks.
- Engineered edge-control scoring, adjacency-constrained titan movement, surround-based elimination, dual timer, single-player bot, undo/redo, and move-history replay.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, TypeScript

Frameworks & Libraries: React, Node.js, Express, FastAPI, Tailwind CSS, Socket.io, LangGraph

Databases: Neo4j, Qdrant, PostgreSQL (Supabase), MongoDB, Redis

Developer Tools: Git, Docker, Docker Compose

Certifications: Fundamentals of Object-Oriented Programming (NPTEL)

ACADEMIC ACHIEVEMENTS

Olympiads: International Rank 10 – IEO | International Rank 5 – IMO | Zonal Rank 3 – NSO

School: School Rank 1 in Class X (AISSE) | School Rank 3 in Class XII (AISSCE)

Mathematics: Top 10% in the inter-level screening test, Association of Mathematics Teachers of India (AMTI)